# TSM 098 Arithmetic Test

Practice 3 | Version 202610122020.3

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

| QUESTIONS | ANSWERS |
|---|---|
| 1. Evaluate at x = -4, y = 5. Show the substituted expression.<br>$5x - 6y - 4$ | 1. Value<br>20 pts |
| 2. Evaluate the same expression at (a) x = 10, (b) x = -10, and (c) x = 0. Show substitution.<br>$3x^2 - 5x + 4$ | 2. (a)<br>7 pts<br>(b)<br>7 pts<br>(c)<br>6 pts |
| 3. Evaluate each expression.<br>$(a) \quad -10 - 7$<br>$(b) \quad (-8)(-6)$<br>$(c) \quad 11 - (-7)$<br>$(d) \quad \frac{9}{0}$ | 3. (a)<br>5 pts<br>(b)<br>5 pts<br>(c)<br>5 pts<br>(d)<br>5 pts |
| 4. Reduce each fraction to lowest terms.<br>$(a) \quad \frac{21}{35}$<br>$(b) \quad \frac{70}{110}$ | 4. (a)<br>10 pts<br>(b)<br>10 pts |
| 5. Evaluate each side separately at x = 4, -4, 0. Record the left value, right value, and whether they are equal. State what the tested values establish; do not solve the equation.<br>$x(x + 4) = x^2 + 4x + 4$ | 5. x = 4: left / right / equal?<br>4 pts<br>x = -4: left / right / equal?<br>4 pts<br>x = 0: left / right / equal?<br>4 pts<br>Conclusion<br>8 pts |

# Instructor answer key: TSM 098 Arithmetic Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

**TEST PAGE 1**

1. Value

$-54$

20 pts

2. (a)

254 — 7 pts

(b)

354 — 7 pts

(c)

4 — 6 pts

3. (a)

-17 — 5 pts

(b)

48 — 5 pts

(c)

18 — 5 pts

(d)

Undefined — 5 pts

4. (a)

$\frac{3}{5}$ — 10 pts

(b)

$\frac{7}{11}$ — 10 pts

5. x = 4: left / right / equal?

Left 32; right 36; no — 4 pts

x = -4: left / right / equal?

Left 0; right 4; no — 4 pts

x = 0: left / right / equal?

Left 0; right 4; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts